

Curriculum Vitae

Patrick E. Carlson

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Education

Iowa State University

Ames, Iowa

PhD, Cumulative GPA: 3.68/4.0

2008 - 2015

Major: Human-Computer Interaction (HCI)

Simpson College

Indianola, Iowa

Bachelor of Arts, cum laude, Cumulative GPA: 3.715/4.0

2003 - 2008

Majors: Computer Science and Psychology

Experience

Sandia National Labs

Albuquerque, New Mexico

Data Scientist (promoted to Principal level - May 2021)

2017 - present

Data analysis, R and Python programming, data virtualization, Tableau and Shiny visualizations, analytics team co-lead, SQL and Mongo querying, architecture and data engineering research, project planning, writeups, design mockups, and UX reviews.

Renaissance Learning

Wisconsin Rapids, Wisconsin

User Experience Analyst

2016 - 2017

Perform remote usability testing on software prototypes, help with UI design, and develop and analyze survey data.

Iowa State University

Ames, Iowa

Improving Open Source Software Development (Dissertation Topic)

2010 - 2015

Apply data mining in the development of algorithms and tools to improve developer understanding of technical and social structure in large Open Source communities. Under direction of Dr. Judy Vance.

Bimanual Haptics for Virtual Assembly Tasks (Research Assistant)

2009 - 2015

Designed and analyzed studies examining various bimanual haptic device configurations and learning transfer for virtual assembly tasks. Under direction of Dr. Judy Vance.

Mentor/Support Research Experience for Undergraduate (REU) Students

Summer 2010

Helped coordinate three undergraduate REU students whose project entailed creating a physical mockup of a shopping cart which was then used in a virtual reality user study. Under direction of Dr. Judy Vance.

University of New Mexico

Albuquerque, New Mexico

Research Experience for Undergraduates (REU)

Spring 2006 - Fall 2007

Learned the basics of machine learning and worked on applying this to improving the routing of packets in a dynamic network. Under direction of Dr. Terran Lane.

Simpson College

Indianola, Iowa

Usability and Appeal of the Linux Desktop (Psychology Capstone)	Spring 2007 - Fall 2008
Constructed and performed an experiment empirically contrasting usability and appeal of a 2D versus 3D Linux desktop.	
Origins of Human Cooperation and Altruism	Fall 2005 - Fall 2006
Simulated the evolution of cooperation/altruism using game theory techniques and genetic algorithms. Presented work at Argonne National Laboratory.	

Student Technician	Nov 2005 - May 2006
Helped students and faculty connect to the Internet and troubleshoot network issues.	

Mounds Park Academy St. Paul, Minnesota

Computer Technician	Summers 2002 - 2005, and 2008
Conducted laptop repair and software setup for 60 freshman students. Imaged and configured multiple computer labs.	

Teaching and Other

Engineering/LAS Online Learning (Graduate Assistant)	Fall 2014
Recorded class videos, advised undergraduate producers, and scripted automated startups for recordings	
ME/WLC 484/584: Technology, Globalization, and Culture (Teaching Assistant)	Fall 2013
Graded homework, facilitated online class discussions	
HCI/Psych 522: Scientific Methods in HCI (Teaching Assistant)	Spring 2013
Graded homework and managed content for online website	
CS 309: Software Development Practices (Teaching Assistant)	Spring 2009
Graded homework, helped students with documenting project development	
CS 207: Programming I (Teaching Assistant)	Fall 2008
Assisted and tutored students with homework and basic programming concepts	

Groups and Activities

Sandia Data Sciences - Community of Practice Organizer	2018 - Present
Iowa State University Badminton Club	2009 - 2015
ISU Badminton Club Webmaster	2009 - 2012
Computational Design Synthesis: Summer Camp (Munich, Germany)	August 1-5, 2011
ISU Human-Computer Interaction Student Group Vice President	2010 - 2011
Simpson College Math Club	2006 - 2008
Simpson College Computer Club	2003 - 2008

Honors

Nominated for Employee Recognition Award (ERA)	March 2026
Sandia Spot Award - Data and AI Governance Framework	October 2025
Sandia Spot Award - Data Sciences Department Transition Excellence	July 2025
Nominated for Lab Director's Award - Sandia AI Chat Team	May 2025
Thunderbird 'Kudos' Award From Fellow Sandian For Poster Preparation	March 2025
Nominated for Employee Recognition Award (ERA) - Hybrid Work Environment	March 2025
Awarded Employee Recognition Award (ERA) - Sandia AI Chat Team	March 2025

Awarded Employee Recognition Award (ERA) - Pioneering Machine Learning and Data Analytics Techniques to Mitigate Lab-wide Risk	March 2025
Sandia Spot Award - Data Architecture Partners	November 2024
Nominated for Employee Recognition Award (ERA) - Risk Analytics: Time Reporting Codes/Time Charging Analytics Team	June 2024
Thunderbird 'Kudos' Award From Fellow Sandian For Data Integration Help	July 2023
Awarded Employee Retention Compensation for Select Employees	June 2023
Thunderbird 'Kudos' Award From Fellow Sandian For Data Visualization Development	March 2023
Nominated for Employee Recognition Award (ERA) - LDRD Demographics Team	March 2023
Thunderbird 'Kudos' Award From Fellow Sandian For Report Recovery	December 2022
Nominated for 'Most Inspiring Up & Comer' for FedScoop 50	September 2022
Sandia Spot Award - Work Agreement Visualization for Senior Leadership Team	September 2022
Thunderbird 'Kudos' Award From Fellow Sandian For Urgent Denodo Help	July 2022
Nominated for Employee Recognition Award (ERA) - Data Integration Team	March 2021
Nominated for Employee Recognition Award (ERA) - Facilities Centralized Space Management Modeling and Analysis Team	March 2020
Iowa State University Research Excellence Award	Spring 2015
Outstanding Senior in Computer Science (Simpson College)	May 2008
Honorable Mention in the Mathematical Contest in Modeling (MCM)	2006, 2007, 2008
Psi Chi National Honor Society in Psychology	March 2007
Simpson College Dean's List	Spring 2005, Fall 2006, Fall 2007
Simpson College Academic Honor Scholarship	Fall 2003 - Spring 2008
Certificate of Achievement from MPA Technology Department	December 2000

Poster Presentations

Carlson, Patrick & Vance, Judy M. (2013, April). [Who Should I Contact?: Helping New Developers Find Experts](#). Poster presented at the Emerging Technologies Conference, Ames, IA. **Awarded most interesting research project.**

Peters, Anicia & Carlson, Patrick & Gilbert, Stephen & Vance, Judy M. (2012, April). [A Hybrid Method to Support Natural Interaction of Parts in a Virtual Environment](#). Poster presented at the Emerging Technologies Conference, Ames, IA. **Awarded most interesting research project.**

Vance, Judy M. & Gilbert, Stephen & Oren, Michael & Pavlik, Ryan & Carlson, Patrick (2012, July). [GOALI: A Hybrid Method to Support Natural Interaction of Parts in a Virtual Environment](#). Poster presented at the NSF Engineering Research and Innovation Conference, Atlanta, Georgia.

Carlson, Patrick & Vance, Judy M. & Nguyen, Tien & Blankenship, Kevin (2011, April). [Social Technical Congruence: The Link Between Social Science and Technology](#). Poster presented at the Emerging Technologies Conference, Ames, IA.

Carlson, Patrick & Vance, Judy M. (2010, May). [An Evaluation of Asymmetric Interfaces for Bimanual Virtual Assembly With Haptics](#). Poster presented at the ASME World Conference on Innovative Virtual Reality, Ames, IA.

Carlson, Patrick (2008, May). Usability and Appeal of a 2D versus 3D Linux Operating System. Poster presented at the Midwestern Psychological Association, Chicago, IL.

Publications

Bruno, Sean & Carlson, Patrick (2023). Podman vs Docker comparison via R Shiny project deployment. In [Proceedings of the 2023 Improving Scientific Software Conference](#), pages 10-14. **(Peer-Reviewed Conference Paper)**

Carlson, Patrick & Vance, Judy M. & Berg, Meisha (2016). An evaluation of asymmetric interfaces for bimanual virtual assembly with haptics. In [Virtual Reality](#), pages 1-9. **(Peer-Reviewed Journal Article)**

Carlson, Patrick & Peters, Anicia & Gilbert, Stephen & Vance, Judy M. & Luse, Andy (2015). Virtual Training: Learning Transfer of Assembly Tasks. In [IEEE Transactions on Visualization and Computer Graphics](#), pages 770-782. **(Peer-Reviewed Journal Article)**

Carlson, Patrick (2015). Engaging developers in open source software projects: harnessing social and technical data mining to improve software development. In [Graduate Theses and Dissertations](#), pages 1-192, Ames, IA. **(PhD Dissertation)**

Carlson, Patrick & Xiao, Nan (2012). Experience and Recommendations for Distributed Software Development. In [Proceedings of the international conference on software engineering \(ICSE\) workshop on collaborative teaching of globally distributed software development](#), pages 1-4, Zurich, Switzerland. **(Workshop Paper)**

Oren, Mike & Carlson, Patrick & Gilbert, Stephen & Vance, Judy M. (2012). Puzzle Assembly Training: Real World vs. Virtual Environment. In [Proceedings of the IEEE 2012 virtual reality conference](#), pages 1-4, Orange County, California. **(Conference Paper)**

Vance, Judy M. & Gilbert, Stephen B. & Oren, Michael & Pavlik, Ryan & Carlson, Patrick (2011). GOALI: A Hybrid Method to Support Natural Interaction of Parts in a Virtual Environment. In NSF engineering research and innovation conference proceedings, pages 1-4, Atlanta, Georgia. **(Workshop Paper)**

Carlson, Patrick & Kirpes, Carl & Pavlik, Ryan A. & Vance, Judy M. & Yin, Livien & Scott-Cooper, Terrence & Lambert, Troy (2011). Comparison of Single-Wall Versus Multi-Wall Immersive Environments to Support a Virtual Shopping Experience. In [Proceedings of the ASME 2011 world conference on innovative virtual reality \(WINVR2011\)](#), pages 1-5, Milan, Italy. **(Conference Paper)**

Invited Presentations

[Governance in a World of Agents and Data - National Laboratories Information Technology Summit - May 5, 2026](#)

[Exploring Text-to-SQL: Impact on Data Warehouses and Data Governance - National Laboratories Information Technology Summit - May 5, 2025](#)

[Got Data? Discovery is the Key - National Laboratories Information Technology Summit - April 11, 2024](#)

[Sandia Insights - A Data Sciences Architecture and Framework - National Laboratories Information Technology Summit - Oct 17, 2022](#)

[Sandia Insights - A Data Sciences Architecture and Framework - National Laboratories Information Technology Summit - May 30, 2019](#)

[Viz Wars: Tableau vs. Shiny - National Laboratories Information Technology Summit - May 29, 2019](#)

[IEEE VR 2015 \(presented by Dr. Judy Vance\), invited by Dr. Ed Swan, talk title: Virtual Training: Learning Transfer of Assembly Tasks - March 23-27, 2015](#)

[IE 681 Cognitive Engineering, invited by Dr. Stephen Gilbert, talk on Open Source communities - December 4, 2014](#)

[HCI 591 Seminar, talk on Open Source socialization - November 16, 2012](#)

Outreach

[Road Less Traveled](#) Spring 2010, 2011, 2012, 2013

Demo of virtual reality system for over 50 middle school and high school girls from across Iowa.

[ISU Badminton Club online tournament support](#) 2010, 2011, 2012, 2014, 2015

Created online tournament registration and administration system.

Professional Organizations

[Special Interest Group on Human-Computer Interaction \(Past Student Member\)](#)

[Association for Computing Machinery \(Past Student Member\)](#)

Conference Attendance

[National Laboratories Information Technology Summit](#) May 5-7, 2026

Kansas City, Missouri - USA

[Red Hat Summit](#) Oct 2, 2025

Minneapolis, Minnesota - USA

[National Laboratories Information Technology Summit](#) May 5-8, 2025

Denver, Colorado - USA

National Laboratories Information Technology Summit Seattle, Washington - USA	April 8-11, 2024
National Laboratories Information Technology Summit Milwaukee, Wisconsin - USA	June 27-30, 2023
National Laboratories Information Technology Summit Albuquerque, New Mexico - USA	Oct 17-19, 2022
National Laboratories Information Technology Summit Boise, Idaho - USA	May 28-31, 2019
RStudio Conference Austin, Texas - USA	Jan 15-18, 2019
Tableau Conference Las Vegas, Nevada - USA	Oct 9-12, 2017
International Conference on Software Engineering (ICSE) Zurich - Switzerland	June 2-9, 2012
IEEE Virtual Reality Conference Costa Mesa, California - USA	March 4-8, 2012
The Association for the Advancement of Artificial Intelligence Vancouver, British Columbia - Canada	July 22-26, 2007
The Association for the Advancement of Artificial Intelligence Boston, Massachusetts - USA	July 16-20, 2006

Training

GitOps: Continuous Delivery on Kubernetes with Flux (LFS269) This course covered Flux and automating the deployment of applications to Kubernetes via Helm and Git-Ops principles.	Sept 10, 2026
Enterprise AI Machine Learning Using Cloudera This course covered generative AI, large language models, vector databases, and application development within Cloudera Machine Learning.	Oct 28-31, 2024
Cloudera Apache Iceberg Training This course covered Apache Iceberg, data snapshots, versioning, and more within Cloudera services.	Aug 5-8, 2024
Cloudera Data Governance Training This course covered the Atlas data catalog and Ranger security and access control services in Cloudera.	Sept 18-19, 2023
Cloudera Spark Developer Training This course covered Spark, Hive, HDFS, Cloudera Machine Learning, and more.	Sept 11-14, 2023
Cloudera Streaming Analytics Training This course covered Flink and Kafka streaming technologies.	Aug 21-23, 2023
Cloudera Data Science Training This course covered Spark, Cloudera Data Science workbench, supervised and unsupervised learning, and more.	Feb 14-17, 2022
Scaled Agile Framework (SAFe) Product Owner / Product Manager Training This training covered the Scaled Agile Framework for project management, requirements gathering, estimating work, and so on.	Jan 11-13, 2022

[Coursera: Machine Learning](#)

August 14, 2020

Online course on Machine Learning taught by Stanford professor Andrew Ng.

[Apache Spark Programming \(DB 105\)](#)

Sept 24-26, 2019

This course covered the fundamentals of Apache Spark including Spark's architecture and internals, the core APIs for using Spark, SQL and other high-level data access tools, as well as Spark's streaming capabilities and machine learning APIs.

[Intermediate Shiny \(RStudio Conference 2019\)](#)

Jan 15-16, 2019

This covered the basics of Shiny, reactivity, modules, and best-practices.

[Intro to Deep Learning](#)

August 2018 - 8 hours

This was an internal course offered at Sandia National Lab that covered deep learning methods.

[Docker Fundamentals and Docker for Enterprise Developers](#)

Feb 19-22, 2018

This course covered Docker fundamentals, Docker swarm, Docker compose files, YAML format, and more.

[R and Shiny Training](#)

April 9-12, 2018

Taught by Matt Pickard this covered the basics of R, dplyr, Shiny, and reactivity.

Technical Skills

Programming Languages: [Python](#), [Lua](#), [Java](#), [R](#), [C#](#), [HTML](#), [Javascript](#), [SQL](#), [PHP](#), [LaTeX](#), [Cypher \(Neo4j\)](#)

Tools: [Git](#), [Subversion](#), [Windows](#), [Linux](#), [CMake](#), [Eclipse](#), [Jupyter](#), [Docker](#), [Travis-CI](#), [Github Actions](#), [Gitlab Runners](#), [Ansible](#), [Terraform](#), [Bicep](#), [Helm](#), [Kubernetes](#)

Programs: [Inkscape](#), [Lyx](#), [Gimp](#), [JustInMind](#), [Balsamiq](#), [UserZoom](#), [Mendeley](#), [Pencil](#), [Tableau](#), [RStudio](#), [Axure](#), [Tibco Data Virtualization](#), [Collibra](#), [yEd](#), [Denodo Data Virtualization](#)

Libraries: [Pandas](#), [SciPy](#), [scikit-learn](#), [jQuery](#), [D3.js](#), [Django](#), [Bootstrap](#), [NodeJS](#), [Shiny](#), [Jekyll](#), [Spark](#), [Tidyverse](#)